Bright Sefah | Computer Science & Engineering

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brightsefah

Education

University of Mines and Technology (UMaT) **Ghana**
BSc Computer Science and Engineering *2022–2026 (Expected)*

- Focus areas: Robotics, Artificial Intelligence, IoT Systems, and Embedded Systems
- Active leadership in multiple technical communities and student organizations

Research Interests

Robotics and Autonomous Systems, Artificial Intelligence, Internet of Things (IoT), Assistive Technologies, Embedded Systems, Computer Vision, and Human-Computer Interaction with focus on developing context-aware solutions for African communities

Technical Skills

Programming: Python, C/C++, JavaScript, Java

Embedded & IoT: ESP32, ESP8266, Arduino, Raspberry Pi, Sensors (TDS, pH, Ultrasonic, IMU), GSM/GPS modules, PCB design fundamentals

AI & Data: Machine Learning, Data Exploration and Analysis, Computer Vision

Robotics: Autonomous navigation systems, Electronics design and prototyping, Sensor integration

Tools & Frameworks: Git, Linux, Docker, Qt Framework, Fusion 360, Blynk, Firebase, Django, LaTeX

Professional Experience

Orcta Technologies **Ghana**
Founding Team Member and Robotics Lead *2024–Present*

- Leading robotics development initiatives and embedded systems projects
- Designing and implementing intelligent robotic solutions for industrial applications

AI Fire Safety Project
Backend Developer *Oct 2024–Oct 2025*

- Developed backend infrastructure for AI-driven fire detection and safety system
- Implemented real-time data processing pipelines and alert mechanisms
- Integrated machine learning models with backend services for automated fire detection

UHuB (Innovation Hub) **UMaT**
Personnel Intern *Oct–Nov 2024*

- Supported innovation programs and entrepreneurship initiatives
- Assisted in organizing tech workshops and innovation challenges

Agri-IoT Laboratory **UMaT**
Research Intern *Oct–Nov 2024*

- Developed IoT solutions for agricultural monitoring and automation
- Worked on sensor networks for precision agriculture applications

Amalitech Internship Boot Camp
Cloud Practitioner Trainee *Oct–Nov 2024*

- Completed intensive training in cloud computing fundamentals
- Gained hands-on experience with cloud infrastructure and services

SokoAerial **Ghana**
Robotics Intern *Oct–Dec 2023*
- Worked on aerial robotics systems and drone technology
- Assisted in prototyping and testing autonomous flight systems

Aaenics Engineering & Robotics Lab
Engineering & Robotics Tutor *Sept–Oct 2024*
- Delivered technical training on robotics fundamentals and embedded systems
- Mentored students in electronics design and programming

Aaenics Festive Kids International Boot Camp
Robotics Instructor *Sept–Oct 2023*
- Taught robotics and programming to young students
- Designed hands-on curriculum for introducing STEM concepts

Selected Projects

Assistive Smartwatch: **Gesture-Based Emergency Alert System for Physically Challenged Individuals**
- Designed wearable device with gesture recognition for emergency communication
- Implemented GPS tracking and GSM-based alert system for caregiver notification
- Integrated multiple sensors (accelerometer, gyroscope) for gesture detection

Blind Navigation: **AI-Enhanced Navigation Aid for Visually Impaired Users**
- Developed ESP32-based mobility assistance device with ultrasonic obstacle detection
- Implemented fall detection algorithm and automatic SMS alert system
- Integrated haptic feedback and audio cues for navigation guidance

IoT Monitoring: **Real-Time Water Quality Monitoring System**
- Built multi-parameter sensor system measuring TDS, pH, EC, and salinity
- Developed WiFi-enabled remote monitoring dashboard with cloud data logging
- Designed alert system for water quality threshold violations

Multi-Sensor Wearable: **Advanced Wearable Computing Platform**
- Integrated gesture control, fitness tracking, environmental sensing, and tilt alarm
- Developed custom firmware for real-time sensor fusion and data processing
- Designed low-power architecture for extended battery life

Browser Extension: **GrabnGo Productivity Tool**
- Created cross-platform browser extension for note-taking and tab management
- Implemented YouTube video download functionality and link organization system

Qt Application: **Bluetooth Device Controller**
- Developed desktop application for Bluetooth device discovery and pairing
- Implemented Arduino-based LED control system with wireless communication

Competitions & Honors

UMaT 8th Innovation & Career Fair Hackathon
1st Position *2024*

ACSES Week Hackathon
1st Position *2024*

Disability Inclusive Hackathon (DIHack)
3rd Position *2023*

UMaT 7th Innovation Fair Hackathon
4th Position *2023*

ACity GDG Hackathon **Team TIL**
Participant *2025*

Aaenics UMaT Team – 9th Innovation & Career Fair Hackathon
Coach *2025*

Leadership & Professional Service

Technical Leadership

IEEE R8 UMaT Student Branch Chapter
Vice Chair *2026*
Leading initiatives to promote engineering excellence and professional development

Google Developer Group (GDG) UMaT
IoT & Robotics Lead/Organizer *2025*
Organizing workshops, hackathons, and technical training sessions in IoT and robotics

Amalitech Coding Club **UMaT**
Vice President *2024–2025*
Coordinating coding workshops and software development training programs

Inclusive Tech Group (ITG) – DIHack
Organizer *2025*

Aaenics Robotics Club Progression

Aaenics Robotics Club **UMaT**
Tutor *2023*

Aaenics Robotics Club **UMaT**
Assistant Electronics Tutor *2024*

Aaenics Robotics Club **UMaT**
President *2025*

Aaenics Robotics Club **UMaT**
Presidential Advisor *2026*

Mentorship

World Robot Olympiad (WRO)
Mentor *Jan 2025–Present*

Aaenics UMaT Team at Ghana Robotics Competition (GRC)
Mentor *May–Oct 2025*

Additional Leadership

Ghana Institution of Engineering (GhIE)
Member, Project Management Committee *Year 2*

Institute of Engineering and Technology (IET) Ghana UMaT Chapter
Member

Extracurricular Leadership

UMaT Mass Choir
Member *2022*

UMaT Mass Choir
Organizer *2024*

UMaT Mass Choir
Deputy Publicity Officer *2025*

UMaT Mass Choir
Vice President *2026*

Professional Memberships

Institute of Electrical and Electronics Engineers (IEEE)
Institute of Engineering and Technology (IET)
Ghana Institution of Engineering (GhIE)
Google Developer Groups (GDG)

References

Available upon request